# CyberScanner SaaS — Production-Ready Security Scanning API

A scalable cybersecurity scanning platform built with FastAPI, Celery, PostgreSQL, and Redis.

## Architecture

```
cyberscanner/
├── app/
│   ├── api/v1/endpoints/    # Route handlers (auth, scan, report, history)
│   ├── core/                # Config, security, logging, rate limiting
│   ├── db/                  # Database engine, session, migrations
│   ├── models/              # SQLAlchemy ORM models
│   ├── schemas/             # Pydantic request/response schemas
│   ├── services/            # Business logic (nmap, zap, scan orchestrator)
│   ├── tasks/               # Celery async task definitions
│   └── utils/               # Helpers (parsers, validators, formatters)
├── tests/
├── scripts/                 # DB init, seed scripts
├── docker/                  # Dockerfiles
├── docker-compose.yml
├── alembic.ini
└── pyproject.toml
```

## Features

- **JWT Authentication** — Access + refresh tokens, bcrypt password hashing
- **Rate Limiting** — Per-user sliding window via Redis (slowapi)
- **Async Task Queue** — Celery + Redis broker for long-running scans
- **Nmap Integration** — Port scanning, service detection, OS fingerprinting
- **OWASP ZAP Integration** — Active/passive web vulnerability scanning
- **Structured JSON Reports** — Normalized findings with CVSS-like severity
- **PostgreSQL Storage** — Full scan history, findings, and audit logs
- **Security Hardened** — CORS, input sanitization, secrets in env vars

## Quick Start

```bash
```

```
# 1. Copy and configure environment
cp .env.example .env

# 2. Start all services
docker-compose up -d

# 3. Run DB migrations
docker-compose exec api alembic upgrade head

# 4. Create first admin user
docker-compose exec api python scripts/create_admin.py
```

## API Endpoints

| Method | Path | Description |
|---|---|---|
| POST | `/api/v1/auth/register` | Register new user |
| POST | `/api/v1/auth/login` | Login, receive JWT |
| POST | `/api/v1/auth/refresh` | Refresh access token |
| POST | `/api/v1/scan/` | Start a new scan |
| GET | `/api/v1/scan/{scan_id}` | Get scan status |
| DELETE | `/api/v1/scan/{scan_id}` | Cancel a running scan |
| GET | `/api/v1/report/{scan_id}` | Full structured report |
| GET | `/api/v1/report/{scan_id}/export` | Export as JSON/PDF |
| GET | `/api/v1/history/` | Paginated scan history |
| GET | `/api/v1/history/stats` | Aggregated stats |

## Environment Variables

See `.env.example` for all required configuration.